the motion of a baseball in flight, but the usual path—the one single path predicted by Newton's laws—contributes *much* more than do all other paths combined. For large objects, it turns out that classical paths are, by an enormous amount, the dominant contribution to the averaging process and so they are the ones we are familiar with. But when objects are small, like electrons, quarks, and photons, many different histories contribute at roughly the same level and hence all play important parts in the averaging process.

You might finally ask: What is so special about the act of observing or measuring that it can compel all the possible histories to ante up, merge together, and yield a single outcome? How does our act of observing somehow tell a particle it's time to tally up the histories, average them out, and commit to a definite result? Why do we humans and equipment of our making have this special power? Is it special? Or might the human act of observation fit into a broader framework of environmental influence that shows, quantum mechanically speaking, we aren't so special after all? We will take up these perplexing and controversial issues in the latter half of this chapter, since not only are they pivotal to the nature of quantum reality, but they provide an important framework for thinking about quantum mechanics and the arrow of time.

Calculating quantum mechanical averages requires significant technical training. And understanding fully how, when, and where the averages are tallied requires concepts that physicists are still working hard to formulate. But one key lesson can be stated simply: quantum mechanics is the ultimate prochoice arena: every possible "choice" something might make in going from here to there is included in the quantum mechanical probability associated with one possible outcome or another.

Classical and quantum physics treat the past in very different ways.

Pruning History

It is totally at odds with our classical upbringing to imagine one indivisible object—one electron or one photon—simultaneously moving along more than one path. Even those of us with the greatest of self-